Mohsen Taheri Shalmani

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EDUCATION

(2020 – 2024) Ph.D. in IT, Mathematics and Physics, University of Stavanger.
(2018 – 2020) MSc in Mathematics and Physics, University of Stavanger.
(2007 – 2009) MSc in Applied Mathematics, Azad University, Science and Research Branch.
(2002 – 2006) BSc in Mathematics, Azad University, Tehran Central Branch.

WORK EXPERIENCE

(2025 – Pres.) AI researcher in **EAST Project**, Kristiania University of Applied Sciences, Oslo.
(2025 – Pres.) Lecturer at the University of Stavanger (20% position).
(2020 – 2024) Doctoral researcher and a member of **BMDLab**, University of Stavanger.
(2015 – 2018) Math and computer teacher, Mojtama Fani, Tehran.
(2010 – 2015) Bank clerk and IT consultant, Karafarin Bank, Tehran.
(2009 – 2010) Freelance math and computer teacher, Tehran.
(2006 – 2007) Math teacher, Sadr High School, Tehran.

SKILLS

Statistical Modeling and Simulation | R, Python, and MATLAB programming | Mathematical Modeling | AI and ML | Deep Learning | Image Processing and LLM via TensorFlow and Transformers | Big Data | PySpark | Data Mining | Differential Equations | Numerical Analysis | Time Series | Signal Processing | Object Modeling | 3D Slicer | Functional Data Analysis | Language Processing | Computational Geometry| Optimization | Neuroimaging | Object-Oriented Programming | Java, Kotlin, C++, C# | System Testing | Containerization | Docker | Database Validation | Web Applications | React | JavaScript | Power BI | Excel| Cloud Computing | Database Programming | Azure | SQL |
GitHub: https://github.com/MohsenTaheriShalmani

Open-Source Contributions

- **EvoMaster** – Contributing to advancing the EvoMaster open-source automated testing tool by equipping it with AI technologies to enhance API testing, improving reliability, security, and automated generation of test cases for web and enterprise systems.
- **Elliptical Tubes** – Development of the **ETRep R package** and **Elliptical Tube Interactive App** designed for statistical shape analysis of tubular structures, enabling object modeling, simulation, and mean shape computation of elliptical tubes for applications in biomedical and geometric data analysis.
- **SlicerSALT** – Contributor in developing hypothesis testing procedures for shape models within the SlicerSALT open-source platform, which provides advanced tools for statistical shape analysis, visualization, and morphometric measurements for biomedical research.

LANGUAGES

English (Fluent) | Norwegian (Intermediate) | Persian (Native).

SUPERVISION

Co-supervised master's thesis**:** *"Statistical Shape Analysis of Caudate Nucleus in Parkinson's Disease"* (2025) by Thomas Eikemo Tjensvold, University of Stavanger. https://hdl.handle.net/11250/3209793

PUBLICATIONS

- Taheri Shalmani, M., Arcuri, A., Galeotti, J.P., Golmohammadi, A. *"Online Machine Learning Prediction of Failed Calls to Enhance REST API Fuzzing"* Accepted for the 19th International Conference on the Quality of Information and Communications Technology (QUATIC 2026)
- Arcuri, A., Zhang, M., Seran, S., Galeotti, J.P., Golmohammadi, A., Garrett, P., Sahin, O., Castagna, F., Kertusha, Taheri Shalmani, M., I. and Susilo, F.F., 2026. "*EvoMaster at REST League 2026 Tool Competition"*. In 19th Search-Based and Fuzz Testing Workshop (SBFT'26).
- Pizer, S. M., Liu, Z., Zhao, J., Tapp-Hughes, N., Damon, J., Zhang, M., Marron, J. S., Taheri Shalmani, M., & Vicory, J. (2025). *"Interior object geometry via fitted frames*". Journal of Mathematical Imaging and Vision. https://doi.org/10.1007/s10851-025-01266-6
- Taheri Shalmani, Mohsen (2025) *"Deriving Spines of Tubular Objects via Deep Learning.* (Preprint) https://doi.org/10.13140/RG.2.2.14238.45126
- Pizer, S. M., Marron, J., Taheri, M., & Vicory, J. *"Object Correspondence for Shape Statistics"*. (2024–25) Interactions of Statistics and Geometry (ISAG) II.
- Taheri Shalmani, Mohsen, Stephen M. Pizer, and Jörn Schulz. (2025) [R package] ETRep *"Analysis of Elliptical Tubes Under the Relative Curvature Condition"*. The Comprehensive R Archive Network (CRAN). https://doi.org/10.32614/CRAN.package.ETRep
- Taheri Shalmani, Mohsen. *"Shape Statistics via Skeletal Structures."* (2024) Ph.D. Thesis, University of Stavanger, Norway. https://doi.org/10.13140/RG.2.2.34500.23685
- Taheri Shalmani, M., Pizer, S. M., & Schulz, J. (2024). *"The Mean Shape under the Relative Curvature Condition"*. Journal of Computational and Graphical Statistics. https://doi.org/10.1080/10618600.2025.2535600
- Taheri, Mohsen, Stephen M. Pizer, and Jörn Schulz. (2023) *"Fitting the Discrete Swept Skeletal Representation to Slabular Objects."* (Under review) https://doi.org/10.48550/arXiv.2409.04079
- Taheri, Mohsen, and Jörn Schulz. (2022) *"Statistical Analysis of Locally Parameterized Shapes."* Journal of Computational and Graphical Statistics 32, no. 2: 658-670. https://www.tandfonline.com/doi/full/10.1080/10618600.2022.2116445
- Liu, Z., Schulz, J., Taheri, M., Styner, M., Damon, J., Pizer, S., & Marron, J. S. (2023). *"Analysis of joint shape variation from multi-object complexes"*. Journal of Mathematical Imaging and Vision, 65(3). https://link.springer.com/article/10.1007/s10851-022-01136-5
- Pizer, Stephen M., J. S. Marron, James N. Damon, Jared Vicory, Akash Krishna, Zhiyuan Liu, and Mohsen Taheri. (2022) *"Skeletons, object shape, statistics."* Frontiers in Computer Science 4: 842637. https://www.frontiersin.org/articles/10.3389/fcomp.2022.842637/full
- Taheri Shalmani, Mohsen. (2020) *"Statistical shape analysis of brain structures."* Master's thesis, University of Stavanger. https://doi.org/10.13140/RG.2.2.12270.42567

REFERENCES

- Dr. Jörn Schulz (UiS) Email: jorn.schulz@uis.no
- Dr. Jan Terje Kvaløy (UiS) Email: jan.t.kvaloy@uis.no
- Dr. Stephen M. Pizer (UNC) Email: smp@cs.unc.edu
- Dr. Andrea Arcuri (Kristiania University) Email: andrea.arcuri@kristiania.no